

Journal of Oral Microbiology

ISSN: 2000-2297 (Online) Journal homepage: www.tandfonline.com/journals/zjom20

Oral microbiome diversity and composition in inflammatory bowel disease patients with periodontitis

Weiwei Zheng , Mengli Xue , Quan Zhou , Jinni Xu , Xing Yu , Xiaohui Kang , Xia Liu & Xue'e Chen

To cite this article: Weiwei Zheng , Mengli Xue , Quan Zhou , Jinni Xu , Xing Yu , Xiaohui Kang , Xia Liu & Xue'e Chen (2026) Oral microbiome diversity and composition in inflammatory bowel disease patients with periodontitis, Journal of Oral Microbiology, 18:1, 2692246, DOI: 10.1080/20002297.2026.2692246

To link to this article: https://doi.org/10.1080/20002297.2026.2692246

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

Published online: 03 Jul 2026.

Submit your article to this journal

Article views: 482

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at https://www.tandfonline.com/action/journalInformation?journalCode=zjom20

JOURNAL OF ORAL MICROBIOLOGY
2026, VOL. 18, NO. 1, 2692246
https://doi.org/10.1080/20002297.2026.2692246

Taylor & Francis
by informa

RESEARCH ARTICLE

OPEN ACCESS Check for updates

Oral microbiome diversity and composition in inflammatory bowel disease patients with periodontitis

Weiwei Zheng[a,b,c,d,1], Mengli Xue[a,b,c,d,1], Quan Zhou[e,1], Jinni Xu[f], Xing Yu[a,b,c,d], Xiaohui Kang[g], Xia Liu[a,b,c,d] and Xue'e Chen[a,b,c,d]

[a]Department of Gastroenterology, The First Affiliated Hospital, Fujian Medical University, Fuzhou, Fujian Province, People's Republic of China; [b]Department of Gastroenterology, National Regional Medical Center, Binhai Campus of the First Affiliated Hospital, Fujian Medical University, Fuzhou, Fujian, People's Republic of China; [c]Fujian Clinical Research Center for Liver and Intestinal Diseases, Fuzhou, Fujian Province, People's Republic of China; [d]The First Clinical Medical College of Fujian Medical University, Fuzhou, Fujian Province, People's Republic of China; [e]Department of Public Health, Fuzhou Center for Disease Control and Prevention, Fuzhou, Fujian, People's Republic of China; [f]Fujian Medical University, Fujian, People's Republic of China; [g]Department of Medicine, Division of Rheumatology & Clinical Immunology, Columbia University Irving Medical Center, New York, USA

ABSTRACT

Background: Inflammatory bowel disease (IBD) and periodontitis are chronic inflammatory disorders sharing immune dysregulation and microbiota imbalance. The oral microbiota may influence IBD through the 'oral-gut axis.'

Methods: Using 2009–2010 NHANES data, we performed 1:1 propensity score matching (age and sex), yielding 50 IBD patients and 50 controls. Periodontitis was defined by CDC/AAP criteria; IBD by self-reported physician diagnosis. Weighted multivariable logistic regression assessed the periodontitis–IBD association. Oral microbiome data from 16S rRNA sequencing were analyzed for α/β-diversity, differential taxa (ANCOM-BC, random forest, LEfSe), co-occurrence networks, and correlations with systemic inflammatory markers.

Results: Periodontitis was significantly associated with increased IBD risk (OR = 4.82, 95% CI: 1.15–20.15, $P = 0.031$). The IBD group showed reduced Shannon diversity ($P = 0.045$) without significant β-diversity differences. Differential genera included enrichment of *Curvibacter* and *Cloacibacterium* in IBD, and *Porphyromonas* and *Fusobacterium* in controls. Co-occurrence networks revealed sparser interactions and altered hub genera in IBD. Correlations with inflammatory markers were generally weak.

Conclusion: Periodontitis is independently associated with prevalent IBD. IBD patients exhibit reduced oral microbial diversity and compositional shifts. These findings support the 'oral-gut axis' hypothesis and suggest oral microbiota as potential non-invasive indicators for further validation.

ARTICLE HISTORY
Received 9 February 2026
Revised 16 June 2026
Accepted 16 June 2026

KEYWORDS
Inflammatory bowel disease; oral-gut axis; microbiome; NHANES; epidemiological association

Introduction

Inflammatory bowel disease (IBD) and periodontitis represent two prevalent chronic inflammatory disorders that continue to pose substantial challenges to global public health [1]. IBD principally comprises Crohn's disease (CD) and ulcerative colitis (UC), both of which are marked by persistent, relapsing inflammation of the gastrointestinal tract. Clinically, IBD is manifested by symptoms such as abdominal pain, diarrhoea, unintended weight loss, and an elevated risk of colorectal carcinoma [2]. The disease exerts a profound negative impact on patients' quality of life, and in paediatric populations, it may lead to growth impairment and delayed sexual maturation, thereby imposing considerable burdens on affected families and society at large [3].

Concurrently, periodontitis is a chronic infectious disease initiated by dental plaque biofilm, resulting in progressive destruction of the tooth-supporting tissues and, in advanced cases, eventual tooth loss [4].

CONTACT Weiwei Zheng zhengweiwei@fjmu.edu.cn Department of Gastroenterology, The First Affiliated Hospital of Fujian Medical university No 20, ChaZhong Road, Taijiang district, Fuzhou, Fujian Province, 35005, People's Republic of China
[1] Weiwei Zheng, Mengli Xue, and Quan Zhou are all co-first authors.

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

Severe periodontitis affects more than one billion individuals globally, with its incidence increasing in parallel with population aging, thus constituting a significant public health concern in the realm of oral health [5,6].

Emerging evidence has elucidated a significant intrinsic association between IBD and periodontitis [7]. Both conditions share common pathophysiological mechanisms, notably immune dysregulation, dysbiosis of the microbiota, and sustained inflammation [8]. The aetiology of IBD is multifactorial, involving genetic predisposition, environmental influences, compromised intestinal barrier integrity, and aberrant immune responses to the intestinal microbiome [9,10]. Analogously, periodontitis is characterised by an exaggerated inflammatory response to periodontal pathogens, such as Porphyromonas gingivalis and *Fusobacterium nucleatum*, fundamentally reflecting an immune imbalance precipitated by microbial dysregulation [11]. Furthermore, the localised inflammatory process in periodontitis can exert systemic effects through the 'oral-systemic connection', leading to the conceptualisation of the 'oral-gut axis'. This concept underscores the premise that perturbations in the oral microbiome may influence the distal gastrointestinal tract via both direct and indirect mechanisms [12–14].

Epidemiological evidence demonstrates a higher prevalence of periodontitis among individuals with inflammatory bowel disease (IBD) compared to the general population, and conversely, IBD is more common in those with periodontitis [8]. Mendelian randomisation analyses have identified Crohn's disease (CD) as a risk factor for periodontitis, while periodontitis appears to exacerbate ulcerative colitis, thereby supporting the concept of an oral-gut axis [15]. Additionally, smoking—an established risk factor for periodontitis—has been linked to increased genetic susceptibility to CD [16]. Nonetheless, the current body of research is characterised by inconsistencies, which may be attributed to limited sample sizes, confounding variables, and heterogeneity in diagnostic criteria. Consequently, further validation through large-scale, population-based studies remains necessary [14].

Recent advances in microbiome research provide novel insights into the mechanisms underlying the oral-gut axis [17]. The oral cavity constitutes the second largest microbial reservoir in the human body and maintains a dynamic homoeostasis with the host under physiological conditions [18]. In periodontitis, pathogenic symbionts—primarily Gram-negative anaerobic bacteria—undergo excessive proliferation, thereby disturbing this equilibrium [19,20]. Oral microorganisms can reach the gastrointestinal tract via two principal routes: continuous ingestion through saliva and hematogenous dissemination from ulcerated periodontal pockets to distant organs, including the intestine [21]. Animal studies have shown that oral pathogens, such as Porphyromonas gingivalis, can aggravate colitis, compromise intestinal barrier integrity, and alter immune homoeostasis [22]. Research by Atarashi K et al. indicates that the oral cavity may function as a reservoir for potential intestinal pathogens that contribute to the exacerbation of gastrointestinal diseases [23]. Furthermore, clinical investigations by Said HS et al. have demonstrated significant alterations in the oral microbiota of IBD patients, including increased abundance of Proteobacteria and decreased counts of Proteobacteria within the salivary microbiome [24]. These findings underscore a pivotal question: Do changes in the oral microbiome serve as an independent risk factor or as a mediating mechanism in the association between periodontitis and IBD? To date, systematic studies integrating clinical phenotypes, comprehensive microbiome profiling, and nationally representative datasets are lacking.

To address these knowledge gaps, the present study used data from the 2009–2010 cycle of the National Health and Nutrition Examination Survey (NHANES), which includes standardised periodontal examinations, self-reported physician-diagnosed inflammatory bowel disease (IBD), and oral microbiome profiling based on 16S rRNA sequencing [25–28]. The study had two primary objectives. First, we aimed to evaluate the cross-sectional association between periodontitis and prevalent IBD after accounting for demographic, socioeconomic, lifestyle, and clinical covariates. Second, we aimed to characterise differences in oral microbial diversity, community composition, and IBD-associated bacterial taxa between participants with and without IBD.

As an exploratory objective, we further assessed whether IBD-associated oral microbial features were accompanied by changes in microbial co-occurrence patterns and whether selected bacterial genera were correlated with systemic inflammatory indices. By integrating population-based periodontal assessment with oral microbiome data, this study sought to clarify the epidemiological and microbial ecological features linking periodontitis, oral dysbiosis, and IBD within the framework of the oral–gut axis.

Materials and methods

Study population

This study utilised data from the National Health and Nutrition Examination Survey (NHANES), a nationally representative database employing a multistage, stratified sampling design. The survey encompasses comprehensive household interviews and physical examinations, aiming to assess the health and nutritional status of the non-institutionalised civilian population in the United States. The NHANES protocol was approved by the Institutional Review Board (IRB) of the National Centre for Health Statistics (NCHS), and informed consent was obtained from all participants prior to their inclusion in the survey. Detailed information regarding the survey design, sampling methodology, and laboratory procedures is available on the NHANES official website (http://www.cdc.gov/nchs/nhanes).

In this study, we analysed data from the 2009–2010 NHANES cycle because this cycle included standardised periodontal examinations, self-reported physician-diagnosed IBD, demographic and clinical covariates, and oral microbiome sequencing data. The initial sample comprised 10,537 participants, of whom 4,961 had available microbiome data. Among these, a relatively small number ($n = 50$) were identified as having IBD. Using a 1:1 propensity score matching strategy based on age and sex, we matched the 50 IBD cases to 50 controls, resulting in a final analytic sample of 100 individuals.

Independent variable

In the NHANES 2009–2012 'Oral Health—Periodontal Examination' protocol, six sites were assessed per tooth, with a maximum of 28 teeth examined per participant. Clinical periodontal measurements comprised two primary indicators: clinical attachment loss (AL) and probing depth (PD). The classification of periodontitis in this study adhered to the criteria established by the Centres for Disease Control and Prevention/American Academy of Periodontology (CDC/AAP). Severe periodontitis was defined as the presence of AL ≥ 6 mm at ≥ 2 interproximal sites (not on the same tooth) and PD ≥ 5 mm at ≥ 1 interproximal site. Participants meeting the criteria for moderate periodontitis were not concurrently classified as having severe periodontitis. Moderate periodontitis was defined as AL ≥ 4 mm at ≥ 2 interproximal sites (not on the same tooth), or PD ≥ 5 mm at ≥ 2 interproximal sites (not on the same tooth). Mild periodontitis was defined as not meeting the aforementioned criteria, but having AL ≥ 3 mm at ≥ 2 interproximal sites, PD ≥ 4 mm at ≥ 2 interproximal sites (not on the same tooth), or PD ≥ 5 mm at one site. Participants who did not fulfil any of these criteria were considered free of periodontitis. In this study, 'periodontitis' encompassed mild, moderate, and severe cases, and participants were stratified according to the presence or absence of periodontitis.

Outcome variable

The definition of inflammatory bowel disease (IBD) in the 2009–2010 National Health and Nutrition Examination Survey (NHANES) was based on self-reported physician diagnosis. Specifically, participants were classified as having IBD if they responded affirmatively to the questionnaire item: 'Have you ever been told by a doctor or other health professional that you have ulcerative colitis or Crohn's disease?'

Sample collection and microbiome data processing

The NHANES research team was responsible for all microbiome assessment procedures and bioinformatics analyses, with data dissemination managed by the National Centre for Health Statistics (NCHS) Research Centre. Our research group undertook the integration and analysis of the relevant datasets. Oral rinse samples were collected by instructing participants to rinse with mouthwash for 5 seconds, followed by three additional rinses of 5 seconds each. Examiners, after donning new gloves, carefully transferred the samples into collection cups, ensuring that the cup rims were not touched to prevent contamination. Genomic DNA was extracted from the oral rinse samples using the Puregene DNA Purification Kit. The purified DNA samples—including NHANES oral rinse samples, blank controls, and quality control samples—were arrayed into 96-well plates and shipped on dry ice to the Knight Laboratory

at the University of California, San Diego. In total, 72 plates were prepared for the 2009–2010 cycle and 60 plates for the 2011–2012 cycle. Although sample allocation across plates was not explicitly randomised, standardised protocols were employed, and quality control samples were included on each plate to minimise potential batch effects. Upon arrival, samples were stored at −20 °C until further processing. PCR amplification and sequencing of the 16S V4 region were performed using 515F-806R primers, resulting in 19 independent HiSeq sequencing runs. In cases where a HiSeq run or lane failed (defined as all samples yielding no reads or fewer than 100 reads), affected samples were re-sequenced in subsequent runs. For samples from the same participant sequenced in multiple runs, only the result from the latest (re-sequenced) run was retained.

Data processing was conducted using QIIME1, which evaluated the demultiplexed raw sequence files, yielding 10,442 forward and 10,440 reverse fastq files. Due to the limited read length of 125 bp, effective paired-end merging was not feasible; thus, only the forward reads were retained for downstream analysis using the DADA2 pipeline (v. 1.2.1). Each sequencing run was processed independently, and results were subsequently merged. This approach generated amplicon sequence variants (ASVs), which represent unique microbial sequences identified in each sample following quality control [26]. ASVs were catalogued in a feature table documenting the read counts per ASV in each sample. A phylogenetic tree was constructed based on the ASV data, and the DADA2 'removeBimeraDenovo' function was employed to filter out unclassified or non-bacterial taxa, thereby facilitating subsequent analyses.

To assess microbial community diversity, α-diversity and β-diversity metrics were calculated using QIIME2. α-diversity, which characterises the diversity within individual samples, was quantified employing four indices: operational taxonomic unit (OTU) richness, Faith's phylogenetic diversity, the Shannon-Wiener index, and the inverse Simpson index. These indices were computed following rarefaction of all samples to a uniform sequencing depth of 10,000 reads per sample. Subsequent analyses revealed a correlation between OTU richness and Faith's phylogenetic diversity, as well as between the Shannon-Wiener and inverse Simpson indices. β-diversity was evaluated using Bray–Curtis and Jaccard distance matrices, which characterise abundance-based and presence/absence-based community dissimilarities, respectively. Principal coordinate analysis (PCoA) was used for visualisation. Between-group differences in beta diversity were assessed using PERMANOVA, and within-group dispersion was evaluated using PERMDISP.

Covariate

Covariates included age (modelled both as a continuous variable and as a categorical variable with groups: < 40, 40–49, 50–59, and 60–69 years), PIR (poverty income ratio; analysed both continuously and categorically as < 1 and ≥ 1), body mass index (BMI; categorised as < 18.5, 18.5–24.9, 25–29.9, and ≥ 30 kg/m^2), gender (male and female), race/ethnicity (Mexican American, Non-Hispanic White, Non-Hispanic Black, Other Hispanic, and Other Race—including Multi-Racial), smoking status (never, former, and current), alcohol consumption (yes or no), physical activity level (high or low), educational attainment (less than high school, high school or equivalent, and college or above), and the presence of diabetes, hypertension, and hyperlipidemia (each as yes or no). Marital status was classified as married, widowed, divorced, separated, never married, or living with partner. All these variables were considered potential confounders and were included in the statistical models.

Statistical analysis

To ensure the national representativeness of the data, we employed the complex sampling design of NHANES, incorporating sampling weights, stratification, and primary sampling units (PSUs). Specifically, the analyses utilised examination weights (WTMEC2YR), stratification variables (SDMVSTRA), and PSUs (SDMVPSU). All statistical analyses were conducted using R software (version 4.4.0). Continuous variables were summarised as weighted means with standard errors, while categorical variables were reported as weighted percentages and frequencies. The association between periodontitis and inflammatory bowel disease (IBD) was assessed using weighted multivariable logistic regression, with periodontitis categorised dichotomously (absence vs. presence). *P*-values for trend and linear trend analyses were calculated to evaluate the consistency of associations. Three models were constructed: (1) an unadjusted model; (2) a model adjusted for sex, race, education, marital status, and age; and (3) a model further adjusted for Age5,

PIR2, BMI5, gender, race, smoking, drinking, exercise, education, hypertension, diabetes, hyperlipidemia, and marital status. Weighted multivariable logistic regression was further performed to explore the potential relationship between periodontitis and IBD. Odds ratios (ORs) for IBD in relation to periodontitis were calculated, and subgroup analyses were conducted according to Age5, PIR2, BMI5, gender, race, smoking, drinking, exercise, education, hypertension, diabetes, hyperlipidemia, and marital status. Statistical significance was defined as a p-value < 0.05.

Microbiome statistical analyses were performed using QIIME2 (v2023.2) and R (version 4.3) [27,28]. The Kruskal–Wallis test was applied to compare alpha diversity indices (Observed ASVs, Shannon, Simpson, and Pielou's evenness) between IBD and non-IBD groups, with significance determined at FDR-adjusted $p < 0.05$. For beta diversity, Bray–Curtis and Jaccard distance matrices were constructed and visualised via principal coordinates analysis (PCoA); inter-group differences were evaluated using PERMANOVA, and within-group dispersion was assessed by PERMDISP. Differentially abundant genera were identified using a 'triple-method' approach: ANCOM-BC, random forest, and LEfSe. Only genera deemed significant by at least two methods and exhibiting consistent effect directions were retained as robust biomarkers. Co-occurrence networks were constructed using SparCC to calculate intra-group genus-level correlation coefficients, with node centrality determined by degree and betweenness centrality metrics. Finally, associations between inflammatory composite indices (dietary inflammatory index (DII), systemic immune-inflammation index (SII), neutrophil-to-lymphocyte ratio (NLR), platelet-to-lymphocyte ratio (PLR), and systemic inflammation response index (SIRI)) and biomarker genera were evaluated using Spearman correlation; associations with $|r| \geq 0.20$ and FDR-adjusted $p < 0.10$ were incorporated into the final network.

Result

Sample features after propensity score matching

Following propensity score matching (PSM), a total of 100 participants were included in the analysis, with 50 individuals in the IBD group and 50 in the non-IBD group. There were no statistically significant differences between the two groups regarding age, PIR, BMI, sex, race, smoking status, alcohol consumption, physical activity level, or major comorbidities (including hypertension, diabetes, and hyperlipidemia), with all P values exceeding 0.05. The prevalence of periodontitis was also comparable between the groups (46.0% vs. 38.0%, $P = 0.418$). However, a significant difference in educational attainment persisted between the groups ($P = 0.037$), as detailed in Table 1.

Univariate logistic regression analysis

The results of the univariate logistic regression analysis indicated that there was no statistically significant association between periodontitis and the prevalence of IBD (OR = 1.39, 95% CI: 0.63–3.08, $P = 0.418$). Similarly, no demographic characteristics, lifestyle factors, or comorbidities demonstrated a statistically significant relationship with IBD prevalence (all $P > 0.05$). Detailed results are presented in Table 2.

Multivariate logistic regression analysis

In the multivariate logistic regression models, the crude model (Model 1) also revealed no significant association between periodontitis and IBD prevalence (OR = 1.39, 95% CI: 0.63–3.08, $P = 0.418$). After adjustment for age, sex, race, educational attainment, and marital status (Model 2), the effect estimate for periodontitis increased but remained statistically non-significant (OR = 2.03, 95% CI: 0.72–5.70, $P = 0.180$). However, after further comprehensive adjustment for socioeconomic status, BMI, lifestyle factors, and comorbidities (Model 3), periodontitis was found to be significantly positively associated with IBD prevalence (OR = 4.82, 95% CI: 1.15–20.15, $P = 0.031$). Full details are provided in Table 3.

Subgroup analysis

Subgroup analyses demonstrated that the direction of the association between periodontitis and IBD was generally consistent across strata defined by age, PIR, BMI, sex, race, smoking/drinking status, physical

Table 1. Comparison of Baseline Characteristics Between IBD and Non-IBD Groups After Propensity Score Matching.

Variable	Total ($n = 100$)	Non-IBD ($n = 50$)	IBD ($n = 50$)	P	SMD
Age, M (Q_1, Q_3)	54.00 (44.00, 63.00)	54.00 (44.50, 62.75)	54.00 (44.50, 62.75)	1.000	0.000
PIR, M (Q_1, Q_3)	2.67 (1.18, 5.00)	3.15 (1.22, 5.00)	2.44 (1.16, 5.00)	0.481	−0.151
BMI, M (Q_1, Q_3)	28.55 (25.53, 31.03)	28.55 (25.69, 30.54)	28.31 (25.38, 31.82)	0.970	0.054
Gender, *n* (%)				1.000	
Male	36 (36)	18 (36.00)	18 (36.00)		0.000
Female	64 (64)	32 (64.00)	32 (64.00)		0.000
Race, *n* (%)				0.503	
Mexican American	24 (24)	9 (18.00)	15 (30.00)		0.262
Other Hispanic	9 (9)	5 (10.00)	4 (8.00)		−0.074
Non-Hispanic White	50 (50)	26 (52.00)	24 (48.00)		−0.080
Non-Hispanic Black	15 (15)	8 (16.00)	7 (14.00)		−0.058
Other Race - Including Multi-Racial	2 (2)	2 (4.00)	0 (0.00)		−0.289
Smoking, *n* (%)				0.963	
Never	44 (44)	22 (44.00)	22 (44.00)		0.000
Former	35 (35)	18 (36.00)	17 (34.00)		−0.042
Now	21 (21)	10 (20.00)	11 (22.00)		0.048
Drinking, *n* (%)				1.000	
No	24 (24)	12 (24.00)	12 (24.00)		0.000
Yes	76 (76)	38 (76.00)	38 (76.00)		0.000
Exercise, *n* (%)				0.221	
Low physical activity	40 (40)	17 (34.00)	23 (46.00)		0.241
High physical activity	60 (60)	33 (66.00)	27 (54.00)		−0.241
Education, *n* (%)				0.037	
Less than high school	31 (31)	17 (34.00)	14 (28.00)		−0.134
high school or equivalent	16 (16)	12 (24.00)	4 (8.00)		−0.590
college or above	53 (53)	21 (42.00)	32 (64.00)		0.458
High blood pressure, *n* (%)				0.546	
No	45 (45)	24 (48.00)	21 (42.00)		−0.122
Yes	55 (55)	26 (52.00)	29 (58.00)		0.122
Diabetes, *n* (%)				0.603	
No	82 (82)	42 (84.00)	40 (80.00)		−0.100
Yes	18 (18)	8 (16.00)	10 (20.00)		0.100
Hyperlipidemia, *n* (%)				0.134	
No	20 (20)	13 (26.00)	7 (14.00)		−0.346
Yes	80 (80)	37 (74.00)	43 (86.00)		0.346
Marital, *n* (%)				0.975	
Married	49 (49)	26 (52.00)	23 (46.00)		−0.120
Widowed	8 (8)	4 (8.00)	4 (8.00)		0.000
Divorced	17 (17)	8 (16.00)	9 (18.00)		0.052
Separated	6 (6)	2 (4.00)	4 (8.00)		0.147
Never married	16 (16)	8 (16.00)	8 (16.00)		0.000
Living with partner	4 (4)	2 (4.00)	2 (4.00)		0.000
Periodontitis, *n* (%)				0.418	
No	58 (58)	31 (62.00)	27 (54.00)		−0.161
Yes	42 (42)	19 (38.00)	23 (46.00)		0.161

Note: Continuous variables are presented as median (Q1, Q3); categorical variables are presented as *n* (%); SMD, standardised mean difference.

activity level, educational attainment, comorbidities, and marital status. Tests for interaction revealed no statistically significant effect modification by any of these variables (all P for interaction > 0.05). Further details are available in Table 4.

Microbiome analysis

Microbial α diversity

Multiple α diversity indices were employed to assess and compare the microbial complexity and evenness between the two groups (Figure 1). There was no significant difference in Observed ASVs (a measure of species richness) between the IBD and No IBD groups (Kruskal–Wallis, $p = 0.420$), indicating that the overall number of detectable taxa was comparable between the cohorts. However, the Shannon index, which accounts for both richness and evenness, revealed a statistically significant difference (Kruskal–Wallis, $p = 0.045$), with the IBD group exhibiting a lower Shannon index, suggestive of reduced community diversity in this group. In contrast, both the Pielou's evenness index ($p = 0.093$) and the Simpson index ($p = 0.121$) demonstrated a non-significant trend, implying that the primary alteration in

Table 2. Univariate Logistic Regression Analysis of Factors Associated with IBD.

Variables	OR (95%CI)	P
Age5		
<40	1.00 (Reference)	
40-49	1.00 (0.28 ~ 3.57)	1.000
50-59	1.00 (0.30 ~ 3.33)	1.000
60-69	1.00 (0.32 ~ 3.10)	1.000
PIR2		
<1	1.00 (Reference)	
> = 1	0.89 (0.34 ~ 2.32)	0.806
BMI5		
<18.5	1.00 (Reference)	
> = 30	0.00 (0.00 ~ Inf)	0.988
18.5-24.9	0.00 (0.00 ~ Inf)	0.988
25-29.9	0.00 (0.00 ~ Inf)	0.988
Gender		
Male	1.00 (Reference)	
Female	1.00 (0.44 ~ 2.26)	1.000
Race		
Mexican American	1.00 (Reference)	
Other Hispanic	0.48 (0.10 ~ 2.27)	0.354
Non-Hispanic White	0.55 (0.20 ~ 1.50)	0.245
Non-Hispanic Black	0.53 (0.14 ~ 1.94)	0.334
Other Race - Including Multi-Racial	0.00 (0.00 ~ Inf)	0.988
Smoking		
Never	1.00 (Reference)	
Former	0.94 (0.39 ~ 2.30)	0.900
Now	1.10 (0.39 ~ 3.11)	0.858
Drinking, *n* (%)		
No	1.00 (Reference)	
Yes	1.00 (0.40 ~ 2.50)	1.000
Exercise		
Low physical activity	1.00 (Reference)	
High physical activity	0.60 (0.27 ~ 1.36)	0.222
Education, *n* (%)		
Less than high school	1.00 (Reference)	
high school or equivalent	0.40 (0.11 ~ 1.54)	0.184
college or above	1.85 (0.76 ~ 4.53)	0.178
High blood pressure		
No	1.00 (Reference)	
Yes	1.27 (0.58 ~ 2.81)	0.547
Diabetes, *n* (%)		
No	1.00 (Reference)	
Yes	1.31 (0.47 ~ 3.66)	0.603
Hyperlipidemia, *n* (%)		
No	1.00 (Reference)	
Yes	2.16 (0.78 ~ 5.98)	0.139
Periodontitis custom		
No	1.00 (Reference)	
Yes	1.39 (0.63 ~ 3.08)	0.418
Marital		
Married	1.00 (Reference)	
Widowed	1.13 (0.25 ~ 5.04)	0.872
Divorced	1.27 (0.42 ~ 3.84)	0.670
Separated	2.26 (0.38 ~ 13.51)	0.371
Never married	1.13 (0.37 ~ 3.50)	0.831
Living with partner	1.13 (0.15 ~ 8.68)	0.906

Abbreviations: OR: Odds Ratio, CI: Confidence Interval.

IBD-associated α diversity may be characterised by a decrease in overall diversity, whereas changes in evenness or dominance are relatively modest or potentially limited by sample size.

Microbial β diversity

Principal coordinate analysis (PCoA) based on distance matrices demonstrated substantial overlap between the two groups in two-dimensional space for Bray–Curtis distances (which reflect differences in community composition and relative abundance; PCoA1 = 26.3%, PCoA2 = 15.8%), with PERMANOVA analysis indicating no significant difference ($p = 0.595$) (Figure 2A). Similarly, analysis using Jaccard distances (which

Table 3. Multivariable Logistic Regression Model for the Association Between Periodontitis and IBD.

Variables	Model1 OR (95%CI)	P1	Model2 OR (95%CI)	P2	Model3 OR (95%CI)	P3
Periodontitis						
No	1.00 (Reference)		1.00 (Reference)		1.00 (Reference)	
Yes	1.39 (0.63 ~ 3.08)	0.418	2.03 (0.72 ~ 5.70)	0.180	4.82 (1.15 ~ 20.15)	0.031

Abbreviations: OR: Odds Ratio, CI: Confidence Interval;
Model1: Crude;
Model2: Adjust: Age, Gender, Race, Education, Marital;
Model3: Adjust: Age, PIR, BMI, Gender, Race, Smoking, Drinking, Exercise, Education, High_blood_pressure, Diabetes, Hyperlipidemia, Marital.

Table 4. Subgroup Analysis and Interaction Test for the Association Between Periodontitis and IBD.

Variables	*n* (%)	No (IBD/Total)	Yes (IBD/Total)	OR (95%CI)	P	*P* for interaction
All patients	100 (100.00)	27/58	23/42	1.39 (0.63 ~ 3.08)	0.418	
Age5						0.219
<40	18 (18.00)	4/12	5/6	10.00 (0.85 ~ 117.02)	0.067	
40–49	20 (20.00)	6/13	4/7	1.56 (0.24 ~ 9.91)	0.640	
50–59	26 (26.00)	7/15	6/11	1.37 (0.29 ~ 6.53)	0.692	
60–69	36 (36.00)	10/18	8/18	0.64 (0.17 ~ 2.38)	0.506	
PIR2						0.487
< 1	21 (21.00)	5/9	6/12	0.80 (0.14 ~ 4.53)	0.801	
> = 1	79 (79.00)	22/49	17/30	1.60 (0.64 ~ 4.01)	0.311	
BMI5						0.615
<18.5	2 (2.00)	2/2	0/0			
> = 30	30 (30.00)	7/14	9/16	1.29 (0.30 ~ 5.43)	0.732	
18.5–24.9	20 (20.00)	4/12	5/8	3.33 (0.51 ~ 21.58)	0.206	
25–29.9	48 (48.00)	14/30	9/18	1.14 (0.35 ~ 3.68)	0.823	
Gender						0.820
Male	36 (36.00)	9/20	9/16	1.57 (0.42 ~ 5.90)	0.503	
Female	64 (64.00)	18/38	14/26	1.30 (0.48 ~ 3.52)	0.611	
Race, *n* (%)						0.995
Mexican American	24 (24.00)	5/9	10/15	1.60 (0.29 ~ 8.74)	0.587	
Other Hispanic	9 (9.00)	2/5	2/4	1.50 (0.11 ~ 21.31)	0.765	
Non-Hispanic White	50 (50.00)	18/39	6/11	1.40 (0.37 ~ 5.37)	0.624	
Non-Hispanic Black	15 (15.00)	2/4	5/11	0.83 (0.08 ~ 8.24)	0.876	
Other Race - Including Multi-Racial	2 (2.00)	0/1	0/1	1.00 (0.00 ~ Inf)	1.000	
Smoking, *n* (%)						0.701
Never	44 (44.00)	17/34	5/10	1.00 (0.24 ~ 4.10)	1.000	
Former	35 (35.00)	6/14	11/21	1.47 (0.38 ~ 5.72)	0.581	
Now	21 (21.00)	4/10	7/11	2.62 (0.45 ~ 15.31)	0.283	
Drinking, *n* (%)						0.932
No	24 (24.00)	8/17	4/7	1.50 (0.25 ~ 8.84)	0.654	
Yes	76 (76.00)	19/41	19/35	1.37 (0.56 ~ 3.40)	0.490	
Exercise, *n* (%)						0.212
Low physical activity	40 (40.00)	15/25	8/15	0.76 (0.21 ~ 2.77)	0.680	
High physical activity	60 (60.00)	12/33	15/27	2.19 (0.77 ~ 6.18)	0.140	
Education, *n* (%)						0.991
Less than high school	31 (31.00)	4/11	10/20	1.75 (0.39 ~ 7.91)	0.467	
high school or equivalent	16 (16.00)	2/10	2/6	2.00 (0.20 ~ 19.91)	0.554	
college or above	53 (53.00)	21/37	11/16	1.68 (0.48 ~ 5.80)	0.415	
High blood pressure, *n* (%)						0.140
No	45 (45.00)	12/31	9/14	2.85 (0.77 ~ 10.57)	0.117	
Yes	55 (55.00)	15/27	14/28	0.80 (0.28 ~ 2.31)	0.680	
Diabetes, *n* (%)						0.133
No	82 (82.00)	22/51	18/31	1.83 (0.74 ~ 4.50)	0.192	
Yes	18 (18.00)	5/7	5/11	0.33 (0.04 ~ 2.52)	0.287	
Hyperlipidemia, *n* (%)						0.093
No	20 (20.00)	1/9	6/11	9.60 (0.88 ~ 105.16)	0.064	
Yes	80 (80.00)	26/49	17/31	1.07 (0.44 ~ 2.65)	0.877	
Marital, *n* (%)						0.371
Married	49 (49.00)	14/28	9/21	0.75 (0.24 ~ 2.34)	0.620	

Table 4. (*Continued*)

Variables	*n* (%)	No (IBD/Total)	Yes (IBD/Total)	OR (95%CI)	P	*P* for interaction
Widowed	8 (8.00)	3/5	1/3	0.33 (0.02 ~ 6.65)	0.472	
Divorced	17 (17.00)	4/10	5/7	3.75 (0.47 ~ 29.75)	0.211	
Separated	6 (6.00)	1/2	3/4	3.00 (0.08 ~ 107.45)	0.547	
Never married	16 (16.00)	5/12	3/4	4.20 (0.33 ~ 53.12)	0.268	
Living with partner	4 (4.00)	0/1	2/3	231297584.81 (0.00 ~ Inf)	0.998	

Abbreviations: OR: Odds Ratio, CI: Confidence Interval.
Note: 'No/Yes (IBD/Total)' indicates the number of IBD cases/total participants within the corresponding periodontitis stratum; P for interaction, P-value for interaction test.

emphasise presence/absence-based compositional differences; PCoA1 = 17.4%, PCoA2 = 8.5%) did not reveal clear separation between groups, and PERMANOVA was likewise non-significant ($p = 0.854$) (Figure 2B). Collectively, these findings suggest that at the overall community level, the structural differences between IBD and No IBD groups are limited, and are more likely attributable to variations in specific taxa rather than a global community shift. In summary, β diversity analyses indicate no substantial differences in overall community structure between the two groups.

Community composition

At the phylum level, both groups were dominated by Firmicutes, Actinobacteria, Bacteroidetes, and Proteobacteria (Figure 3), with no prominent differences in overall composition. Boxplot distributions further demonstrated that the dominant phyla exhibited similar ranges of abundance across groups.

At the genus level, the dominant genera were highly consistent between groups (Figure 4), with Streptococcus prevailing, accompanied by Rothia, Prevotella (including multiple subtypes/numbered genera), Veillonella, Haemophilus, Gemella, Actinomyces, *Fusobacterium*, and Neisseria as common dominant taxa. Heatmap analyses revealed individual variability among samples, yet the overall group-level patterns remained similar.

Differential analysis

To identify IBD-associated differential taxa and construct potential biomarkers, we employed a cross-validation approach utilising ANCOM-BC, Random Forest (RF), and LEfSe analyses (Figures 5–7).

The volcano plot generated by ANCOM-BC revealed significant differences in the abundance of several genera between the two groups (Figure 5). Specifically, the No IBD group was enriched in genera such as Yersinia, Shuttleworthia, Faecalibacterium, Gardnerella, Anaeroglobus, and an unclassified genus within the family Comamonadaceae. In contrast, the IBD group exhibited higher relative abundances of Curvibacter, Cloacibacterium, and an unclassified genus within the order Burkholderiales. Corresponding bar plots of relative abundance further illustrated both the magnitude and variability of these taxa between groups.

Random Forest analysis was performed to identify discriminative features from a classification perspective (Figure 6). The importance ranking indicated that Porphyromonas, *Fusobacterium*, Dialister, Catonella, and Anaeroglobus were the primary contributors to group differentiation. Notably, Micrococcus, despite its low abundance, emerged as a distinguishing feature between groups. Comparative abundance analysis showed that, with the exception of Micrococcus—which was more abundant in the IBD group—Porphyromonas, *Fusobacterium*, Dialister, Catonella, and Anaeroglobus were generally more prevalent in the No IBD group.

LEfSe analysis yielded results highly consistent with those of the RF model (Figure 7). According to LDA scores, Micrococcus was preferentially enriched in the IBD group, whereas *Fusobacterium*, Porphyromonas, Dialister, Anaeroglobus, and Catonella were more abundant in the No IBD group. The LEfSe-based abundance comparison further corroborated the directionality and magnitude of these intergroup differences for the core genera.

In summary, multiple analytical approaches consistently identified several key genera as robust candidate biomarkers distinguishing IBD from No IBD, underscoring their potential utility in the differential diagnosis of IBD.

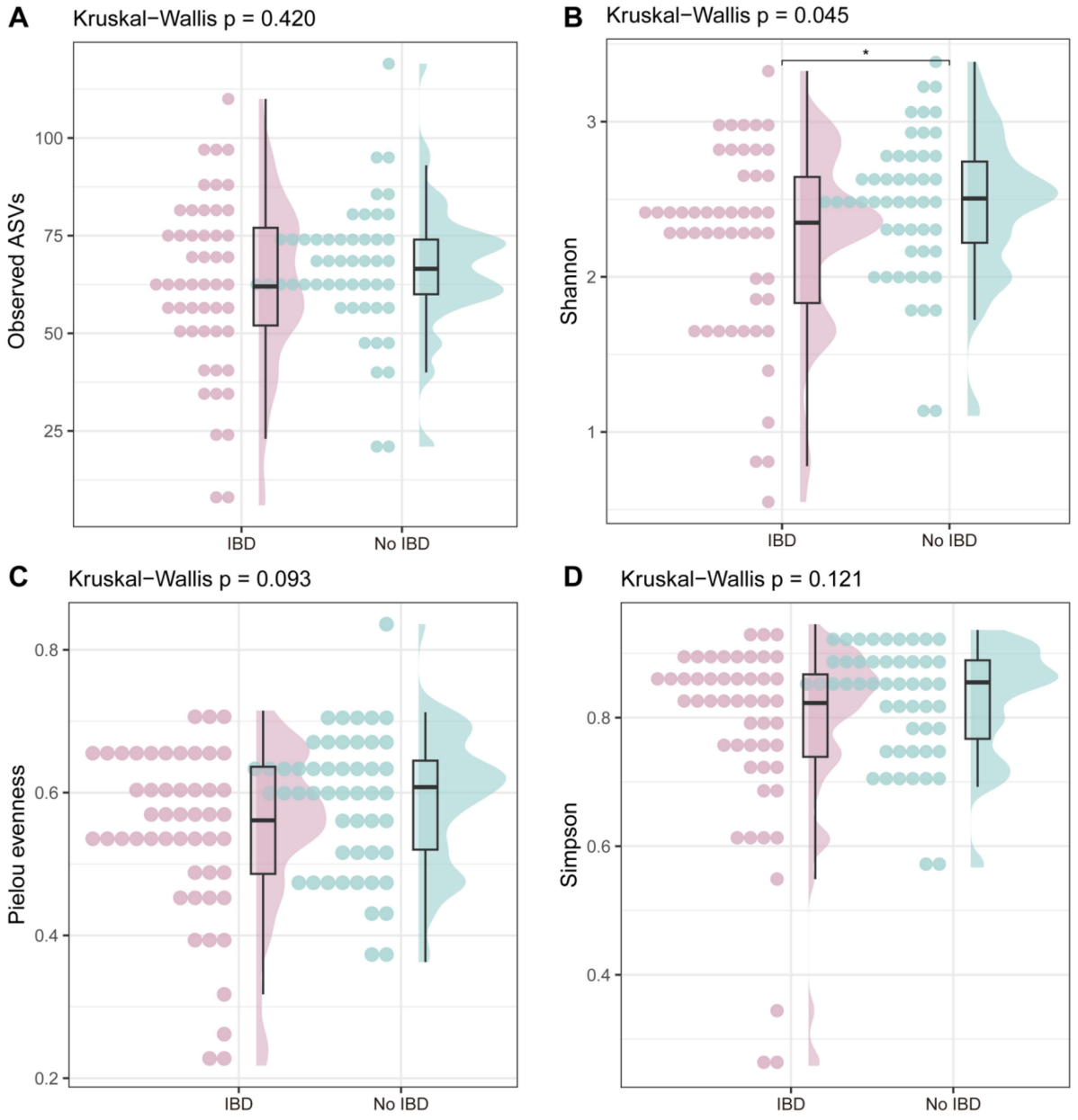


Figure 1. *α*-diversity comparison between IBD and non-IBD groups visualised via raincloud plots (combining half-violin plots, box plots, and jittered individual points). Panels show (A) richness (Observed ASVs), (B) Shannon diversity, (C) Pielou evenness, and (D) Simpson diversity. Statistical significance was determined by Kruskal–Wallis H-test.

Co-occurrence network analysis

Microbial interaction networks within groups were constructed based on SparCC-inferred correlations (Figure 8), revealing pronounced differences in both network architecture and hub taxa between the two cohorts. The network corresponding to the IBD group was notably sparser, characterised by reduced connectivity and complexity. In this group, Capnocytophaga, Haemophilus, and unclassified genera within the order Bacteroidales emerged as central hub nodes, suggesting that these taxa may play pivotal roles in the co-occurrence relationships underlying IBD-associated ecological niches.

In contrast, the No IBD group exhibited a denser and more intricate network, with Prevotella_6, Megasphaera, and Veillonella serving as the principal hubs. This pattern indicates that, in the absence of IBD, microbial community interactions tend to be more stable and diverse, with different dominant anaerobic genera occupying core positions within the network.

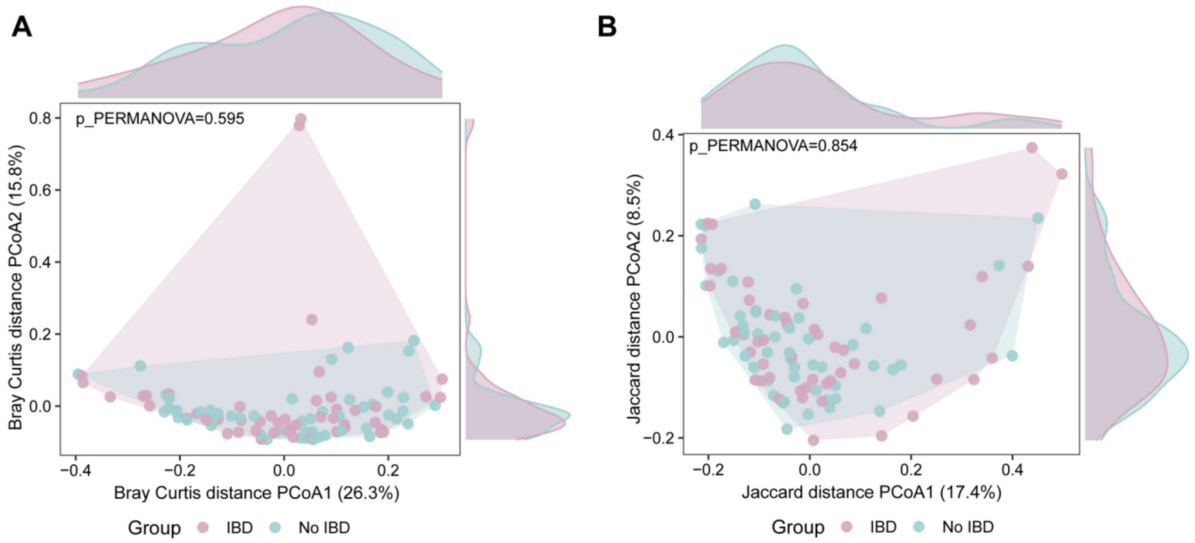


Figure 2. Principal coordinate analysis (PCoA) based on Bray–Curtis dissimilarity and Jaccard distance, with permutational multivariate analysis of variance (PERMANOVA). Left: PCoA of quantitative Bray–Curtis distances; Right: PCoA of presence/absence Jaccard distances. PERMANOVA *P*-values for group differences are shown.

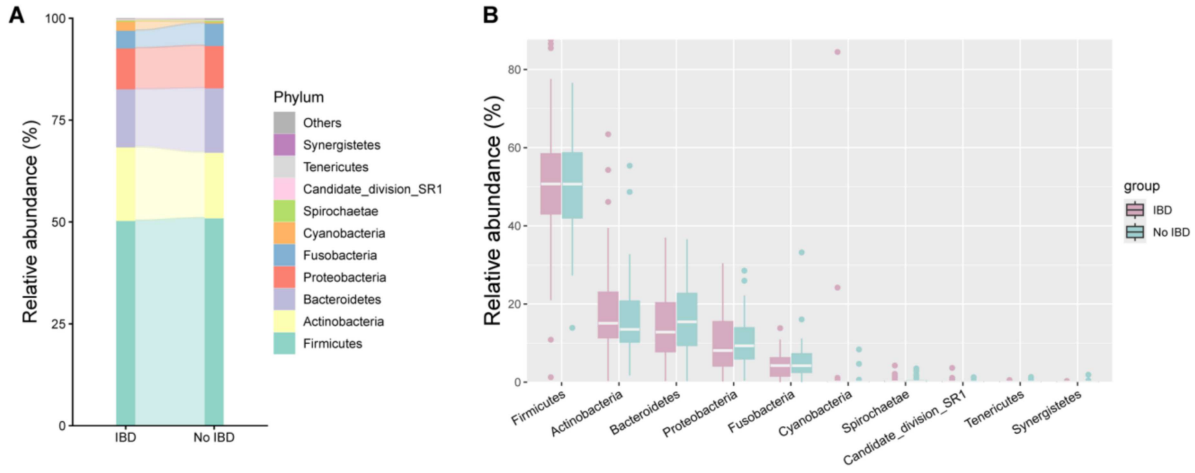


Figure 3. Phylum-level taxonomic composition of the oral microbiota. (A) Stacked bar plot illustrating the relative abundance of major phyla in the overall cohort; (B) Box plots depicting the distribution of the 10 most abundant phyla across individual samples in IBD and non-IBD groups.

Additionally, both positive and negative edges were observed within the networks, indicating the coexistence of potential synergistic and antagonistic relationships in both groups. Notably, the dominant nodes and the overall interaction architecture were restructured between groups, underscoring a group-specific reorganisation of microbial interactions.

Correlation analysis between inflammation-related indices and differential genera

Correlation analyses integrating multiple composite inflammation-related indices (such as DII, SII, NLR, PLR, and SIRI) with differential or signature microbial genera (Figure 9) demonstrated a robust correlation structure among the inflammatory indices themselves, as reflected by moderate to strong inter-index correlations in the heatmap. This finding suggests a high degree of consistency or collinearity in the inflammatory status captured by these metrics within the study cohort.

At the 'index–genus' level, network visualisation indicated that most associations exhibited weak effect sizes or failed to reach statistical significance, with the majority of connections categorised as 'Not' in the

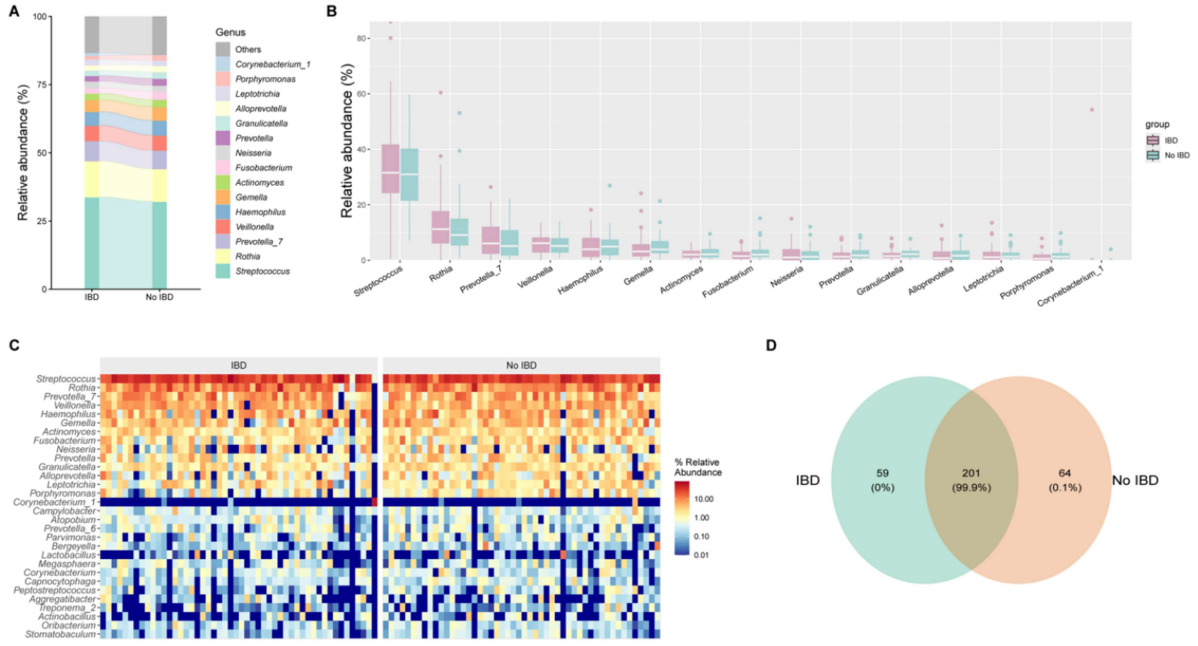


Figure 4. Taxonomic profiling at the genus level. (A) Stacked bars represent the 20 most abundant genera; (B) Box-and-whisker plots for the top 15 discriminatory genera; (C) Heatmap visualisation of genus abundance (Top 30) with sample clustering; (D) Two-set Venn diagram depicting the intersection of genera detected in IBD and control groups.

network. Only a limited number of associations achieved moderate or higher correlation strength ($|Cor| \geq 0.2$). Collectively, these results imply that, under the current data conditions, the correspondence between IBD-associated microbial alterations and systemic inflammatory indices may display certain trends but remains insufficiently robust. Further validation and refinement will require larger sample sizes, stratified adjustments (e.g. for age, sex, medication), and stringent control for multiple comparisons.

Discussion

In this study, key confounding factors were rigorously controlled using propensity score matching (PSM), in conjunction with multidimensional microbiome analyses, to investigate the epidemiological association between periodontitis and inflammatory bowel disease (IBD) as well as the potential oral microbial ecological mechanisms underlying this relationship. After comprehensive adjustment for socioeconomic status, lifestyle variables, and comorbidities, a significant positive association was observed between periodontitis and IBD risk (OR = 4.82, 95% CI: 1.15–20.15, $P = 0.031$). In contrast, this association did not reach statistical significance in models with no adjustment or adjustment for demographic variables alone. These findings suggest the presence of negative confounding suppression, wherein certain factors inversely associated with both periodontitis and IBD may mask or attenuate the true relationship between the two conditions when not adequately controlled. This interpretation is corroborated by a meta-analysis conducted by Wang Z et al., which similarly demonstrated that confounding factors can bias the observed association between periodontitis and IBD [7]. Additionally, the present study achieved preliminary balance in age and sex through PSM and further incorporated variables such as education and poverty income ratio (PIR) in Model 3, thereby enhancing the robustness of the association. Nevertheless, the possibility of residual confounding cannot be excluded. Notably, the 95% confidence interval for the odds ratio in Model 3 was relatively wide (1.15–20.15), reflecting the limited post-matching sample size and consequent uncertainty in the effect estimate. These findings indicate that while the observed association is directionally informative, the precise magnitude of the effect requires validation in larger-scale prospective studies.

A substantial body of literature supports an association between inflammatory bowel disease (IBD) and periodontitis. A recent meta-analysis, consistent with our findings, demonstrated a significantly increased

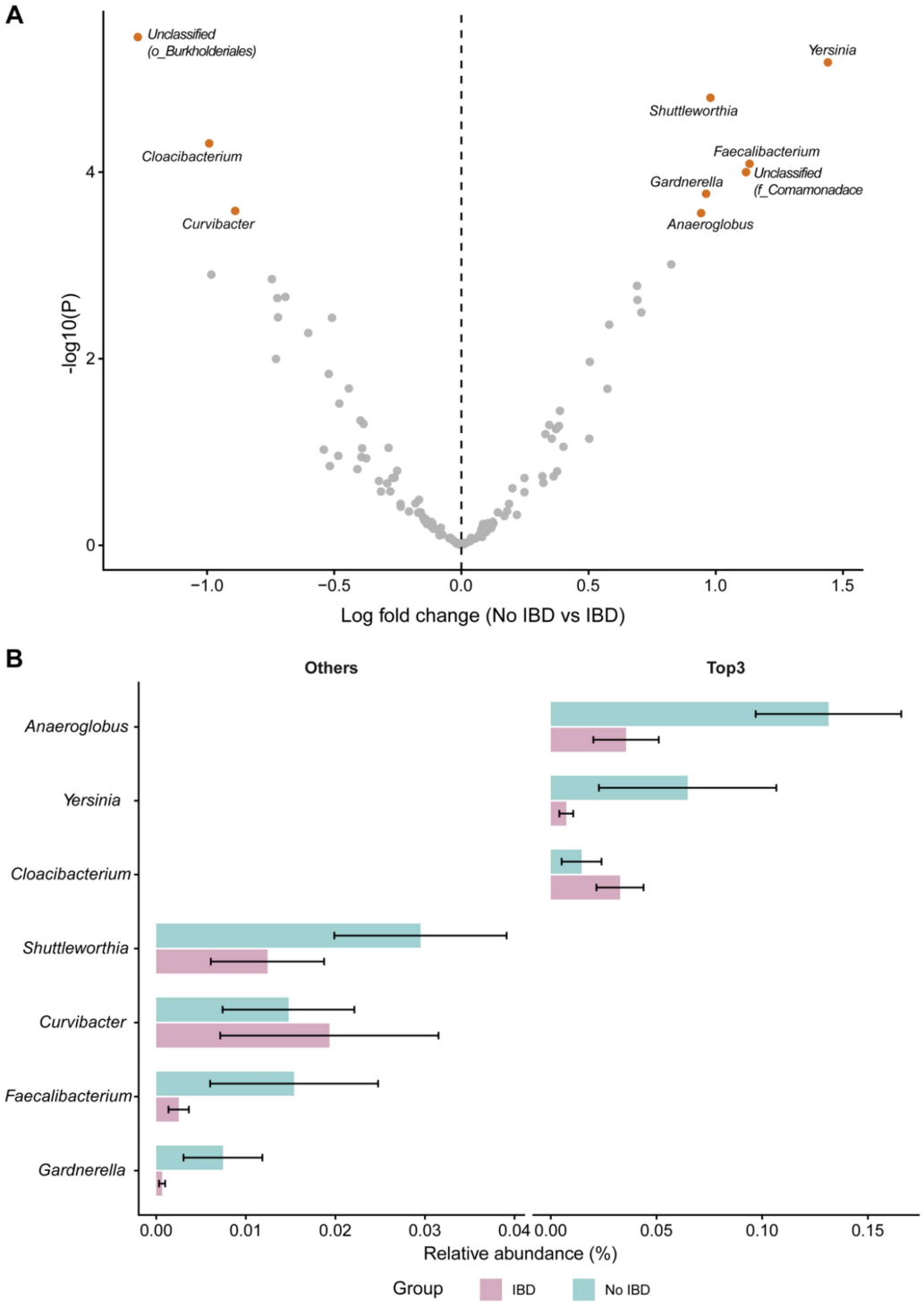


Figure 5. ANCOM-BC identification of differentially abundant genera between IBD and non-IBD groups. (A) Volcano plot: x-axis, log2 fold change (negative values indicate depletion in IBD); y-axis, -log10(adjusted *P* value); labelled points represent significantly altered genera. (B) Relative abundance profiles of significant hits across groups.

overall risk of periodontitis among individuals with IBD [29]. Similarly, She YY et al. conducted a meta-analysis involving 599 IBD patients and 448 controls, revealing a significant association between periodontitis and IBD [30]. Baima et al. further reported that IBD patients are approximately 3.5 times more likely to develop moderate to severe periodontitis compared to controls, thereby reinforcing our conclusions [31]. However, it is noteworthy that the literature remains inconsistent regarding the strength and

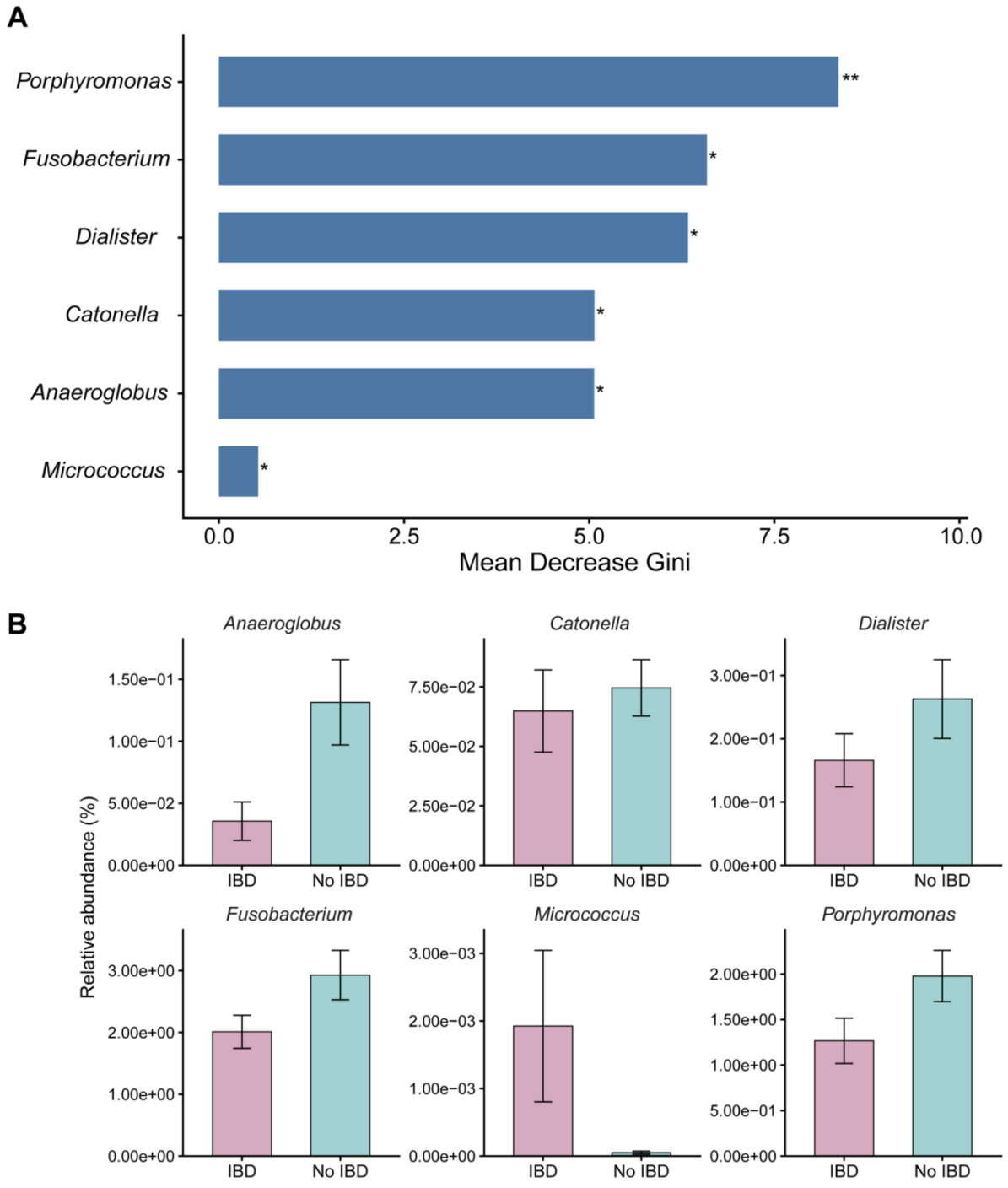


Figure 6. Random forest feature selection of key bacterial genera discriminating IBD from non-IBD. (A) Feature importance plot ranked by Gini importance; (B) Box plots showing relative abundance distributions of top-ranked genera across groups.

direction of this association. For example, a cohort study based on a Korean population found no significant association between periodontitis and Crohn's disease risk [32], while Williams KM et al., analysing data from a U.S. population, reported no significant association between periodontal disease and IBD risk [33]. In addition, a Mendelian randomisation analysis also failed to establish a causal relationship [34]. Such heterogeneity may be attributed to several factors: first, the diagnostic criteria for periodontitis vary considerably across studies, ranging from self-reported data to comprehensive full-mouth examinations, leading to differences in methodological rigour; second, the criteria for diagnosing IBD are not

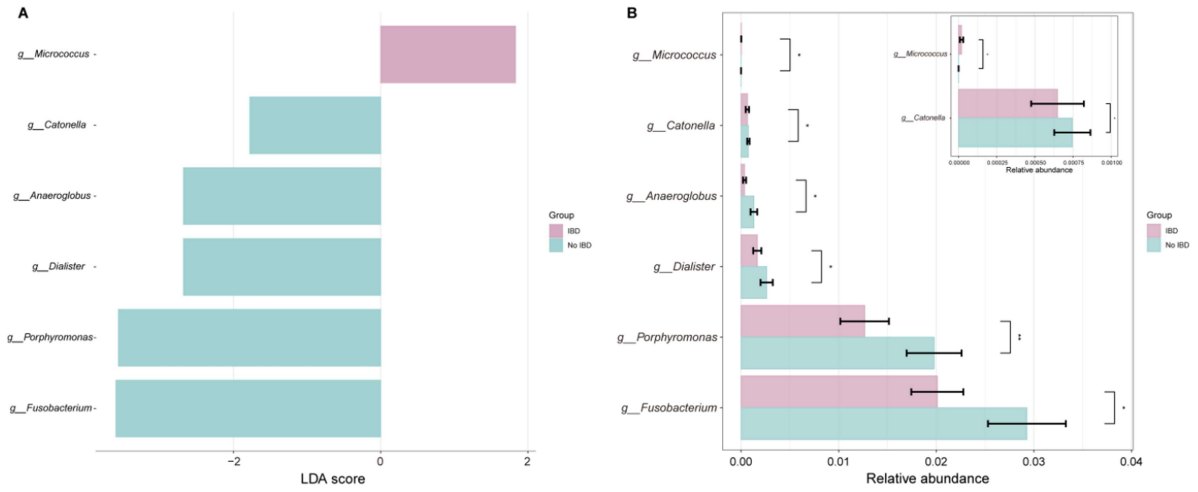


Figure 7. Differential genus analysis using random forest selection and LEfSe (Linear Discriminant Analysis Effect Size). (A) Bar plot of LDA scores (log10 transformed) for discriminative genera with LDA > 2.0; (B) Relative abundance comparison of identified biomarker genera between IBD and non-IBD groups.

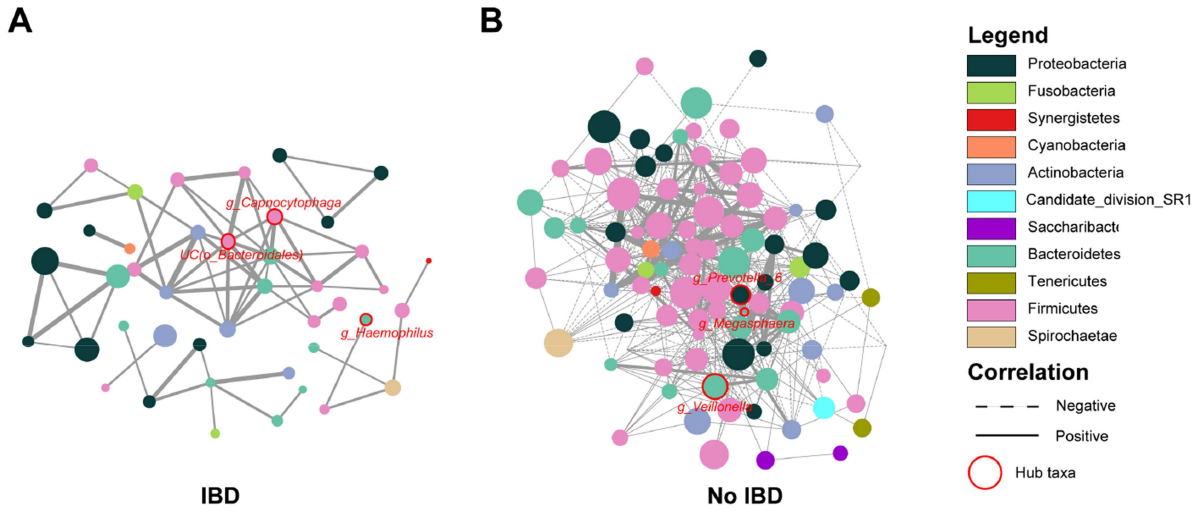


Figure 8. Intra-group microbial association networks for IBD and non-IBD cohorts. (A) Co-occurrence network of the IBD group; (B) Co-occurrence network of the non-IBD group. Nodes represent genera (coloured by phylum); edges represent SparCC correlations (solid: positive, dashed: negative, thickness proportional to correlation magnitude). Hub genera (top 10% by betweenness centrality) are marked in red.

standardised, encompassing self-report, clinical diagnosis, or hospitalisation records; third, the degree of adjustment for confounding variables, particularly strong confounders such as smoking, obesity, and socioeconomic status, differs substantially among studies. In the present study, periodontitis was defined using the standardised CDC/AAP periodontal examination criteria, and propensity score matching (PSM) along with multiple model adjustment strategies were employed, thereby enhancing the reliability and comparability of our results. Furthermore, our findings are largely consistent with previous literature, further substantiating the association between IBD and periodontitis. Nevertheless, the subtle differences in the strength and direction of associations reported across studies underscore the need for more in-depth mechanistic investigations and high-quality prospective evidence to elucidate the causal relationship between these two conditions.

In our microbiome analysis, the Shannon index for α-diversity was significantly reduced in IBD patients, indicating decreased within-community species evenness and complexity. However, β-diversity

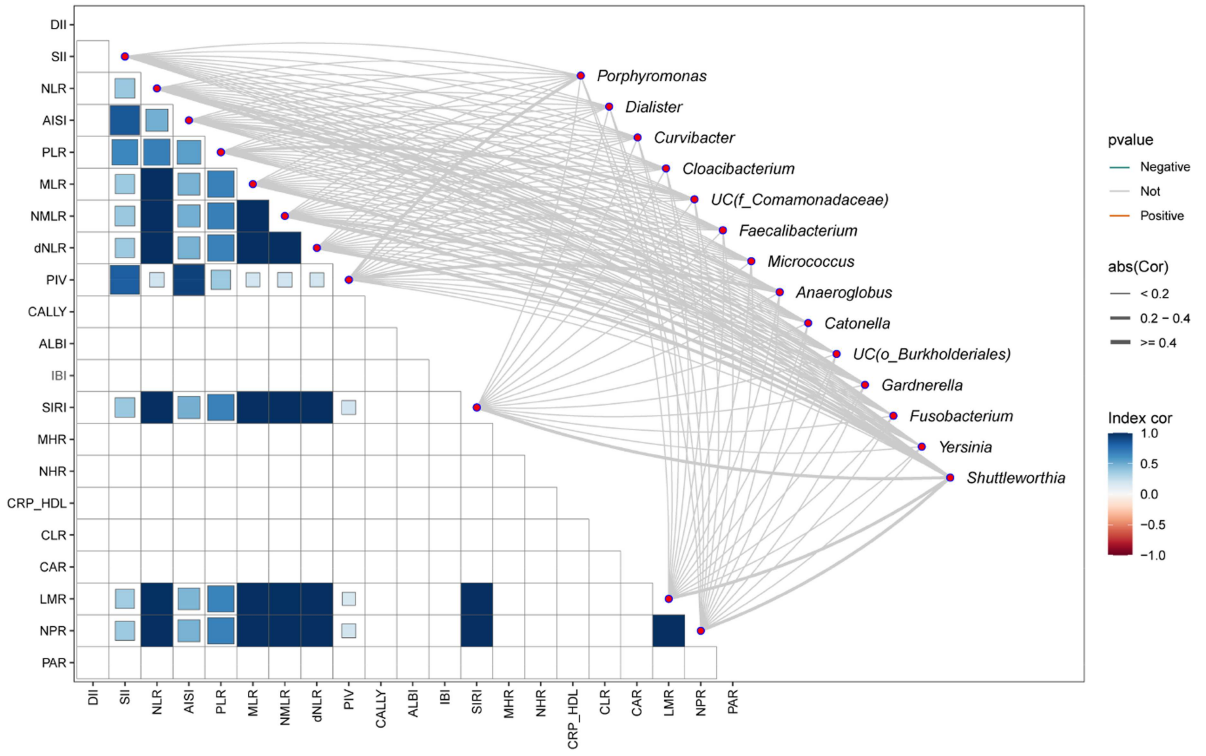


Figure 9. Integration of host inflammatory profiles and microbiome signatures. (A) Spearman correlation heatmap of clinical inflammatory indices (colour scale: blue negative, red positive); (B) Bipartite network linking inflammation indices to key bacterial genera. Edge coloration: blue (negative correlation), red (positive correlation), grey (non-significant); edge width proportional to |correlation coefficient|. Key genera were identified through differential abundance testing (ANCOM-BC/LEfSe) and random forest feature selection.

analyses based on Bray-Curtis and Jaccard distances did not reveal significant overall separation in community structure between groups. These findings suggest that oral dysbiosis in IBD is primarily characterised by diminished intra-community diversity rather than a fundamental or large-scale shift in the overall taxonomic composition. This pattern is not unique to our study; similar trends have been observed in other investigations of microbiota associated with chronic diseases. For instance, a case-control study involving a U.S. cohort demonstrated a significant reduction in oral microbial richness among IBD patients compared to healthy controls [35]. Likewise, Molinero *N* et al. reported a significant decrease in the Shannon diversity index in saliva samples from patients with ulcerative colitis (UC), with β-diversity analysis indicating the formation of two distinct clusters separating UC patients from healthy controls [36]. These results imply that disease states may disrupt the internal dynamic equilibrium of the microbiota, alter interspecies relative abundances, and perturb microbial interaction networks. The proliferation of a limited number of dominant taxa may lead to niche monopolisation and a concomitant reduction in community compositional complexity. Our co-occurrence network analysis corroborated these findings, demonstrating a shift from a previously balanced and diverse microbial network toward a simplified, pathogen-dominated structure under disease conditions, with diminished ecological resilience. Such microecological imbalance is likely to serve as a crucial link mediating the interplay between the two diseases.

Regarding the composition of the microbiota, we identified several characteristic microorganisms enriched in either inflammatory bowel disease (IBD) or periodontitis, which facilitates further inference concerning their pathogenic mechanisms. Numerous pathogenic bacteria are recognised as principal contributors to periodontal tissue destruction; notably, several of these have also been detected as colonising the gut, where they may exacerbate IBD [37,38]. Through LEfSe differential analysis, we observed significant enrichment of *Fusobacterium*, Porphyromonas, Bacteroidaceae, and Enterobacteriaceae in the gut microbiota of patients with concurrent IBD and periodontitis, whereas beneficial taxa such as Bifidobacteriaceae and Lactobacillaceae

were more abundant in healthy controls. The identification of these differential bacterial genera and network hub taxa provides novel insights and candidate targets for the development of microbiome-assisted diagnostic tools aimed at IBD risk prediction, disease stratification, or therapeutic response assessment.

Furthermore, our study revealed a significant correlation between the systemic inflammatory status of IBD and the severity of periodontitis. This finding is consistent with the report by Figueredo et al., who observed markedly elevated expression of pro-inflammatory cytokines, including IL-1β, IL-6, and TNF-α, in the gingival tissues of patients during active IBD compared to those in remission, with these cytokine levels positively correlating with clinical indices of intestinal disease severity [39]. Multiple additional studies have corroborated these observations [40,41]. Notably, research by Pietropaoli et al. also demonstrated a significant association between the severity of periodontal disease and IBD severity [42]. These data suggest that systemic inflammation in IBD may disseminate to periodontal tissues, thereby initiating or exacerbating periodontal inflammatory responses. Previous studies have shown that patients with periodontitis exhibit significantly higher serum levels of LPS, IL-1, and IL-6 compared to healthy controls [43], and these inflammatory mediators are known to promote mucosal immune activation and tissue injury in IBD. Based on these findings, a multi-step mechanistic model can be proposed to explain the potential associations among periodontitis, the oral microbiome, and IBD: firstly, periodontitis facilitates extensive proliferation of oral microbes and production of virulence factors [44]; secondly, these oral bacteria and their products may impact the distal gut via two principal pathways—direct translocation through saliva and indirect immune activation—thereby aggravating or precipitating intestinal inflammation. In this study, we observed reduced oral microbial diversity, an imbalance in the ratio of specific pro-inflammatory and symbiotic taxa, and decreased ecological network stability in IBD patients, which may reflect the oral manifestations of this pathological cascade. Such alterations in the oral microbial ecosystem may continuously deliver aberrant microbial inputs to the gut or sustain a chronic, low-grade systemic inflammatory state, thereby predisposing to the onset or progression of IBD. It should be emphasised that this proposed model is based on observational data and remains hypothetical; definitive causal relationships and underlying molecular mechanisms require validation through experimental and interventional studies.

The strengths of this study are as follows: (1) By integrating epidemiological analyses with high-throughput sequencing technologies, we investigated the association and underlying mechanisms between periodontitis and inflammatory bowel disease (IBD) at both the population level and the microbial community level. In contrast to previous studies that focused solely on epidemiological surveys or laboratory research, our approach provides a comprehensive evidentiary chain encompassing both 'phenomenological associations' and 'biological mechanisms'. (2) Our case group includes both major IBD subtypes—Crohn's disease (CD) and ulcerative colitis (UC)—and is accompanied by detailed clinical data and multiple inflammatory markers. This enables us to explore the influence of IBD subtypes and disease activity on periodontal health, as well as to analyse the role of inflammatory mediators in the comorbidity of these two diseases. (3) We utilised multivariate models to adjust for common confounding factors, thereby minimising confounding bias in the association analyses and enhancing the reliability of our conclusions. (4) In the microbiome analyses, we employed a range of analytical approaches to comprehensively characterise dysbiosis and alterations in microbial interactions, which provides a more systematic and in-depth assessment compared to studies relying on single indicators.

Based on the findings and limitations of the present study, future research may be advanced in the following directions: (1) given the relatively small matched sample size and wide confidence intervals, large-scale prospective cohort studies with long-term follow-up are needed to validate whether periodontitis and specific oral microbial profiles can predict IBD risk; (2) randomised controlled trials evaluating the effects of standardised periodontal therapy on IBD activity, relapse, systemic inflammation, and gut microbiota composition may provide more direct interventional evidence; (3) although α-diversity, β-diversity, taxonomic composition, differential abundance, and co-occurrence network analyses were performed, future studies using metagenomics, metabolomics, metatranscriptomics, and host immunological profiling are needed to better characterise microbial community dynamics and functional pathways along the oral–gut axis; (4) despite propensity score matching and multivariable adjustment, residual confounding cannot be fully excluded, and future studies should collect more detailed data on diet, medication use, antibiotic exposure, oral hygiene behaviours, IBD subtype and activity, and genetic predisposition; and (5) because the correlations between

differential genera and systemic inflammatory markers were generally weak, larger studies integrating repeated inflammatory measurements, paired oral–gut microbiome profiling, and causal inference methods are warranted to clarify the potential mechanistic links among periodontitis, oral dysbiosis, systemic inflammation, and IBD.

Conclusion

In summary, this NHANES-based study provides population-based evidence that periodontitis is independently associated with prevalent IBD after adjustment for multiple potential confounders. In parallel, oral microbiome profiling revealed reduced alpha diversity, alterations in selected bacterial genera, and reorganisation of microbial co-occurrence patterns among participants with IBD. These findings support the relevance of the oral–gut axis and highlight oral microbial dysbiosis as a potential ecological feature associated with IBD. Given the cross-sectional design and limited matched sample size, the findings should be interpreted as association-based and hypothesis-generating. Future large-scale prospective studies integrating detailed clinical phenotyping, deeper microbiome profiling, and host inflammatory measurements are warranted to validate these observations and clarify the biological role of oral microbial alterations in IBD.

Acknowledgements

All authors have read and agreed to the published version of the manuscript.

Author contributions

CRediT: **Weiwei Zheng:** Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing; **Mengli Xue:** Conceptualization, Formal analysis, Software, Visualization, Writing – original draft, Writing – review & editing; **Quan Zhou:** Data curation, Investigation, Software, Validation, Writing – review & editing; **Jinni Xu:** Conceptualization, Formal analysis, Software, Visualization, Writing – original draft, Writing – review & editing; **Xing Yu:** Conceptualization, Formal analysis, Software, Visualization, Writing – original draft, Writing – review & editing; **Xiaohui Kang:** Formal analysis, Validation, Writing – original draft, Writing – review & editing; **Xia Liu:** Formal analysis, Validation, Visualization, Writing – review & editing; **Xue'e Chen:** Data curation, Investigation, Software, Validation, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This study was supported by the Natural Science Foundation of Fujian Province (Nos. 2025J01746 and 2026J001473).

Data availability statement

NHANES datasets are publicly available (https://www.cdc.gov/nchs/nhanes/). Analysis scripts and derived data will be made available from the corresponding author upon reasonable request.

Ethics approval statement

NHANES obtained IRB approval and written informed consent from participants; the present analysis of de-identified public data was deemed exempt. All procedures adhered to the ethical guidelines approved by the Ethics Committee of Fujian Medical University (Approval number: MRCTA, ECFAH of FMU [2022] 317).

References

[1] Hajishengallis G, Chavakis T. Local and systemic mechanisms linking periodontal disease and inflammatory comorbidities. Nat Rev Immunol. 2021;21(7):426–440. doi: 10.1038/s41577-020-00488-6

[2] Di Vincenzo F, Yadid Y, Petito V, et al. Circular and circulating DNA in inflammatory bowel disease: from pathogenesis to potential molecular therapies. Cells. 2023;12(15):1953. doi: 10.3390/cells12151953
[3] Benchimol EI, Bernstein CN, Bitton A, et al. The impact of inflammatory bowel disease in Canada 2018: a scientific report from the Canadian gastro-intestinal epidemiology consortium to Crohn's and colitis Canada. J Can Assoc Gastroenterol. 2019;2(Suppl 1):S1–S5. doi: 10.1093/jcag/gwy052
[4] Kinane DF, Stathopoulou PG, Papapanou PN. Periodontal diseases. Nat Rev Dis Primers. 2017;3:17038. doi: 10.1038/nrdp.2017.38
[5] GBD 2017 Inflammatory Bowel Disease Collaborators, The global, regional, and national burden of inflammatory bowel disease in 195 countries and territories, 1990-2017: a systematic analysis for the global burden of disease study 2017. Lancet Gastroenterol Hepatol. 2020;5(1):17–30. doi: 10.1016/S2468-1253(19)30333-4
[6] Chen MX, Zhong YJ, Dong QQ, et al. Global, regional, and national burden of severe periodontitis, 1990-2019: an analysis of the global burden of disease study 2019. J Clin Periodontol. 2021;48(9):1165–1188. doi: 10.1111/jcpe.13506
[7] Wang Z, Li S, Tan D, et al. Association between inflammatory bowel disease and periodontitis: a bidirectional two-sample mendelian randomization study. J Clin Periodontol. 2023;50(6):736–743. doi: 10.1111/jcpe.13782
[8] Zhan C, Zhou Z, Huang Y, et al. Exploration of the shared gene signatures and molecular mechanisms between periodontitis and inflammatory bowel disease: evidence from transcriptome data. Gastroenterol Rep (Oxf). 2023;11:goad041. doi: 10.1093/gastro/goad041
[9] Graham DB, Xavier RJ. Pathway paradigms revealed from the genetics of inflammatory bowel disease. Natur. 2020;578(7796):527–539. doi: 10.1038/s41586-020-2025-2
[10] Ni J, Wu GD, Albenberg L, et al. Gut microbiota and IBD: causation or correlation? Nat Rev Gastroenterol Hepatol. 2017;14(10):573–584. doi: 10.1038/nrgastro.2017.88
[11] Tanwar H, Gnanasekaran JM, Allison D, et al. Unravelling the oral-gut axis: interconnection between periodontitis and inflammatory bowel disease, current challenges, and future perspective. J Crohns Colitis. 2024;18(8):1319–1341. doi: 10.1093/ecco-jcc/jjae028
[12] Genco RJ, Sanz M. Clinical and public health implications of periodontal and systemic diseases: an overview. Periodontol 2000. 2020;83(1):7–13. doi: 10.1111/prd.12344
[13] Hajishengallis G. Interconnection of periodontal disease and comorbidities: evidence, mechanisms, and implications. Periodontol 2000. 2022;89(1):9–18. doi: 10.1111/prd.12430
[14] Ancona A, Petito C, Iavarone I, et al. The gut-brain axis in irritable bowel syndrome and inflammatory bowel disease. Dig Liver Dis. 2021;53(3):298–305. doi: 10.1016/j.dld.2020.11.026
[15] Qing X, Zhang C, Zhong Z, et al. Causal association analysis of periodontitis and inflammatory bowel disease: a bidirectional mendelian randomization study. Inflamm Bowel Dis. 2024;30(8):1251–1257. doi: 10.1093/ibd/izad188
[16] Georgiou AN, Ntritsos G, Papadimitriou N, et al. Cigarette smoking, coffee consumption, alcohol intake, and risk of Crohn's disease and ulcerative colitis: a mendelian randomization study. Inflamm Bowel Dis. 2021;27(2):162–168. doi: 10.1093/ibd/izaa152
[17] Wang A, Zhai Z, Ding Y, et al. The oral-gut microbiome axis in inflammatory bowel disease: from inside to insight. Front Immunol. 2024;15:1430001. doi: 10.3389/fimmu.2024.1430001
[18] Ma Y, Jiang Z, Wang Y, et al. Tongue coating microbiota-based machine learning for diagnosing digestive system tumours. J Oral Microbiol. 2025;17(1):2487645. doi: 10.1080/20002297.2025.2487645
[19] Hajishengallis G, Darveau RP, Curtis MA. The keystone-pathogen hypothesis. Nat Rev Microbiol. 2012;10(10):717–725. doi: 10.1038/nrmicro2873
[20] Santi-Rocca J, Martín-García DF, Lorca-Alonso I, et al. Microbial complexes in subgingival plaque: a bacterial meta-taxonomic study. J Clin Periodontol. 2025;52(7):983–998. doi: 10.1111/jcpe.14138
[21] Imhann F, Bonder MJ, Vich Vila A, et al. Proton pump inhibitors affect the gut microbiome. Gut. 2016;65(5):740–748. doi: 10.1136/gutjnl-2015-310376
[22] Arimatsu K, Yamada H, Miyazawa H, et al. Oral pathobiont induces systemic inflammation and metabolic changes associated with alteration of gut microbiota. Sci Rep. 2014;4:4828. doi: 10.1038/srep04828
[23] Atarashi K, Suda W, Luo C, et al. Ectopic colonization of oral bacteria in the intestine drives TH1 cell induction and inflammation. Sci. 2017;358(6361):359–365. doi: 10.1126/science.aan4526
[24] Said HS, Suda W, Nakagome S, et al. Dysbiosis of salivary microbiota in inflammatory bowel disease and its association with oral immunological biomarkers. DNA Res. 2014;21(1):15–25. doi: 10.1093/dnares/dst037
[25] Zipf G, Chiappa M, Porter KS, et al. National health and nutrition examination survey: plan and operations, 1999-2010. *Vital and health statistics. Ser. 1, Programs and collection procedures*. 2013;(56):1–37.
[26] Callahan BJ, Mcmurdie PJ, Rosen MJ, et al. DADA2: high-resolution sample inference from illumina amplicon data. Nature Methods. 2016;13(7):581–583. doi: 10.1038/nmeth.3869
[27] Bolyen E, Rideout JR, Dillon MR, et al. Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. Nat Biotechnol. 2019;37(8):852–857. doi: 10.1038/s41587-019-0209-9
[28] Lin H, Peddada SD. Analysis of compositions of microbiomes with bias correction. Nat Commun. 2020;11(1):3514. doi: 10.1038/s41467-020-17041-7

[29] Zhang Y, Qiao D, Chen R, et al. The association between periodontitis and inflammatory bowel disease: a systematic review and meta-analysis. BioMed Res Int. 2021;2021:6692420. doi: 10.1155/2021/6692420
[30] She YY, Kong XB, Ge YP, et al. Periodontitis and inflammatory bowel disease: a meta-analysis. BMC Oral Health. 2020;20(1):67. doi: 10.1186/s12903-020-1053-5
[31] Baima G, Muwalla M, Testa G, et al. Periodontitis prevalence and severity in inflammatory bowel disease: a case-control study. J Periodontol. 2023;94(3):313–322. doi: 10.1002/JPER.22-0322
[32] Kang EA, Chun J, Kim JH, et al. Periodontitis combined with smoking increases risk of the ulcerative colitis: a national cohort study. World J Gastroenterol. 2020;26(37):5661–5672. doi: 10.3748/wjg.v26.i37.5661
[33] Williams KM, Challa PK, Lopes EW, et al. Periodontal disease is not associated with risk of inflammatory bowel disease: results from two prospective cohort studies in the US. Aliment Pharmacol Ther. 2023;58(10):1052–1061. doi: 10.1111/apt.17732
[34] Wang Y, Zhou Y, Song F, et al. Unraveling the association between apical periodontitis and inflammatory bowel disease through mendelian randomization. J Endod. 2025;51(8):1044–1052.e13. doi: 10.1016/j.joen.2025.05.008
[35] Kang S-B, Kim H, Kim S, et al. Potential oral microbial markers for differential diagnosis of crohn's disease and ulcerative colitis using machine learning models. Microorganisms. 2023;11(7):1665. doi: 10.3390/microorganisms11071665
[36] Molinero N, Taladrid D, Zorraquín-Peña I, et al. Ulcerative colitis seems to imply oral microbiome dysbiosis. Curr Issues Mol Biol. 2022;44(4):1513–1527. doi: 10.3390/cimb44040103
[37] Kuziak A, Heczko P, Pietrzyk A, et al. Iron homeostasis dysregulation, oro-gastrointestinal microbial inflammatory factors, and Alzheimer's disease: a narrative review. Microorganisms. 2025;13(1):122. doi: 10.3390/microorganisms13010122
[38] Wang A, Zhai Z, Ding Y, et al. The oral-gut microbiome axis in inflammatory bowel disease: from inside to insight. Front Immunol. 2024;15:1430001. doi: 10.3389/fimmu.2024.1430001
[39] Figueredo CM, Martins AP, Lira-Junior R, et al. Activity of inflammatory bowel disease influences the expression of cytokines in gingival tissue. Cytokine. 2017;95:1–6. doi: 10.1016/j.cyto.2017.01.016
[40] Byrd KM, Gulati AS. The "Gum-Gut" axis in inflammatory bowel diseases: a hypothesis-driven review of associations and advances. Front Immunol. 2021;12:620124. doi: 10.3389/fimmu.2021.620124
[41] Tanwar H, Gnanasekaran JM, & Allison D, et al. Unraveling the Link between Periodontitis and Inflammatory Bowel Disease: Challenges and Outlook. Preprint. ArXiv. 2023;arXiv:2308.10907v1. Published 2023 Aug 19. 2023.
[42] Pietropaoli D, Del Pinto R, Corridoni D, et al. Occurrence of spontaneous periodontal disease in the SAMP1/YitFc murine model of crohn disease. J Periodontol. 2014;85(12):1799–1805. doi: 10.1902/jop.2014.140316
[43] Sakai F, Ozasa K, Shimizu K, et al. The potential link between abdominal migraine and chronic apical periodontitis: a case report. Cureus. 2025;17(1):e78270. doi: 10.7759/cureus.78270
[44] Dahlen G, Basic A, Bylund J. Importance of virulence factors for the persistence of oral bacteria in the inflamed gingival crevice and in the pathogenesis of periodontal disease. J Clin Med. 2019;8(9):1339. doi: 10.3390/jcm8091339